

Outdoor Rock Climbing — Gear, Grading, and Safety Basics

Climbing Disciplines

Outdoor rock climbing is generally divided into bouldering, which is done without ropes on shorter routes under 6 meters with crash pads for protection; sport climbing, which uses pre-placed bolted anchors and a rope; and traditional (trad) climbing, where the climber places and removes their own protective gear, such as cams and nuts, as they ascend.

The Yosemite Decimal System

In the United States, route difficulty is measured using the Yosemite Decimal System (YDS), ranging from 5.0 for the easiest roped climbs to 5.15 for the hardest routes in the world as of current records. Grades are further refined with letters a through d at the higher end, for example 5.12b or 5.13c.

Essential Gear

A standard sport climbing rack includes a dynamic climbing rope, typically 60-70 meters long and 9.5-10.2mm in diameter; a climbing harness rated to hold falls; a helmet to protect against falling rock; 12-15 quickdraws to clip the rope into bolted anchors; and a belay device to control rope friction.

Belaying Basics

Belaying is the technique of managing rope tension to catch a climber in the event of a fall. The belayer must keep a hand on the brake side of the rope at all times. A standard belay device such as an ATC or a GriGri creates friction that allows the belayer to hold significantly more force than they could with their grip strength alone.

Fall Factor

Fall factor is calculated by dividing the length of a fall by the length of rope paid out between the climber and the belay device. A fall factor of 2 (the maximum possible) occurs when a lead climber falls before clipping any protection above the belayer, producing the highest possible force on the system.

Weather and Rock Conditions

Climbing on wet rock significantly reduces friction and is considered unsafe on most rock types, particularly sandstone, which can be permanently damaged by climbing while damp. Most outdoor climbing guidebooks recommend waiting at least 24-48 hours after rain before climbing on sandstone.